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Journal of Luminescence 110 (2004) 225–231
JOURNAL OF LUMINESCENCE
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Optical absorption spectra of linear and cyclic thiophenes—selection rules manifestation

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Available online 12 October 2004

Abstract

We theoretically study the size-dependent relation between absorption spectra of thiophene-based oligomers and the corresponding cyclothiophenes. In our approach based on a Frenkel exciton Hamiltonian, we demonstrate that the geometry and selection rules determine the observed relations between the spectra.

PACS: 78.20.Bh; 78.40.Me

Keywords: Oligothiophenes; Cyclothiophenes; Frenkel excitons; Optical absorption

1. Introduction

The dependence of the electronic properties on the chain length of conjugated oligomers and on the size of ring-shaped cyclothiophenes is investigated. Oligomers are well-defined entities where π electrons define the electronic properties of the lowest excited states. For short oligomers their length is assumed to coincide with the delocalization length of the π electrons.

The possibility to synthesize systematically by stepwise assembly of a defined repeat unit oligothiophenes [1,3] with controlled chain length even up to hundred repeat units [4] allows to study size-dependent effects with high precision. Recently using the same thiophene-based repeat units the first series of fully conjugated macrocyclic structures, cyclo(oligothiophene–diacetylenes) and cyclo[*n*]thiophenes, were synthesized [5,6]. Such cyclothiophenes are ring shaped. The absorption spectra for the thiophene oligomers and cyclothiophenes show a systematic redshift with increasing number of thiophene units. The π– π* transition of linear oligothiophenes and ring-shaped cyclothiophenes exhibit an interesting relation between their linear optical absorption spectra. In particular, the

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0022-2313/$ - see front matter
doi:10.1016/j.jlumin.2004.08.013

energy of the absorption band of a cyclothiophene with a given size corresponds to the one of an oligothiophene chain of about half the length. It means that cyclothiophene spectra are blueshifted with respect to oligothiophene spectra with the same number of repeat units.

In the present contribution, we address an approach, based on a Frenkel exciton Hamiltonian, which describes the observed size-dependent redshift of the oligothiophene absorption spectra and the relationship between cyclothiophenes and the corresponding oligothiophenes.

2. Structures and optical properties of the linear oligothiophenes and cyclo[n]thiophenes

Series of monodisperse linear conjugated oligomers, in particular oligothiophenes, in contrast to the corresponding polydisperse polymers possess a defined structure and defined chain and conjugation length. The preparation of several homologous series of oligothiophenes by controlled iterative step-by-step addition of monomeric repeating units allows the detailed investigation of the electronic properties in dependence of the (conjugated) chain length which corresponds to the number of conjugated double bonds [7]. The wealth of the oligomeric approach can be seen in resulting clear structure–property relationships in which typically a linear dependence of the electronic properties (absorption, emission energies, redox potentials) with the (inverse) chain length is found.

In this respect, we recently synthesized a whole series of oligothiophenes (*n*T) ranging from a trimeric thiophene (3T) to a 19-mer (19T) [2,3]. Each second thiophene unit bears at the free β-positions two butyl side chains in order to provide solubility. Therefore, the repeating unit was a bithiophene leading to intermediate chain lengths of 5T, 7T, 9T, 11T, 13T, 15T and 17T. The general structural formula of the *n*T series is given in Fig. 1.

In Table 1 we summarize the absorption data of the oligothiophenes which were measured in dichloromethane. As usual for conjugated oligomer series a progressive redshift and an increasing intensity (see ε value) of the longest wavelength absorption (π–π* transition) with increasing chain length is found. For the longer derivatives the energy differences however become smaller and a beginning saturation is observed.

Recently, we were able to synthesize the first macrocyclic oligothiophenes which represent novel structures in the area of conjugated oligomers and polymers [6]. These cyclo[*n*]thiophenes (C*n*T) contain the same repeating units as the linear series *n*T which therefore is a perfect linear model system for the macrocyclic structures. Derivatives comprising 12 (C12T), 16 (C16T) and 18 (C18T) thiophene units have been synthesized (see Fig. 2).

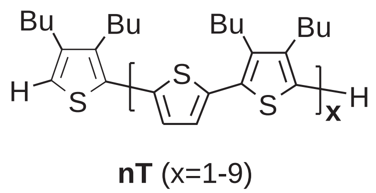

Fig. 1. The general structural formula of the *n*T series (Bu: butyl side chain).

Table 1
Absorption maxima and extinction coefficients of the longest wavelength absorption for oligothiophenes *n*T (Solvent: dichloromethane)

*n*T	3T	5T	7T	9T	11T	13T	15T	17T	19T
$\lambda_{max}^{\pi-\pi^*}$									
(nm)	342	386	409	418	422	425	427	429	430
(eV)	3.625	3.212	3.031	2.966	2.938	2.917	2.904	2.890	2.883
ε(l/mol cm)	18,500	26,600	41,000	55,700	71,500	88,000	100,500	115,000	125,000

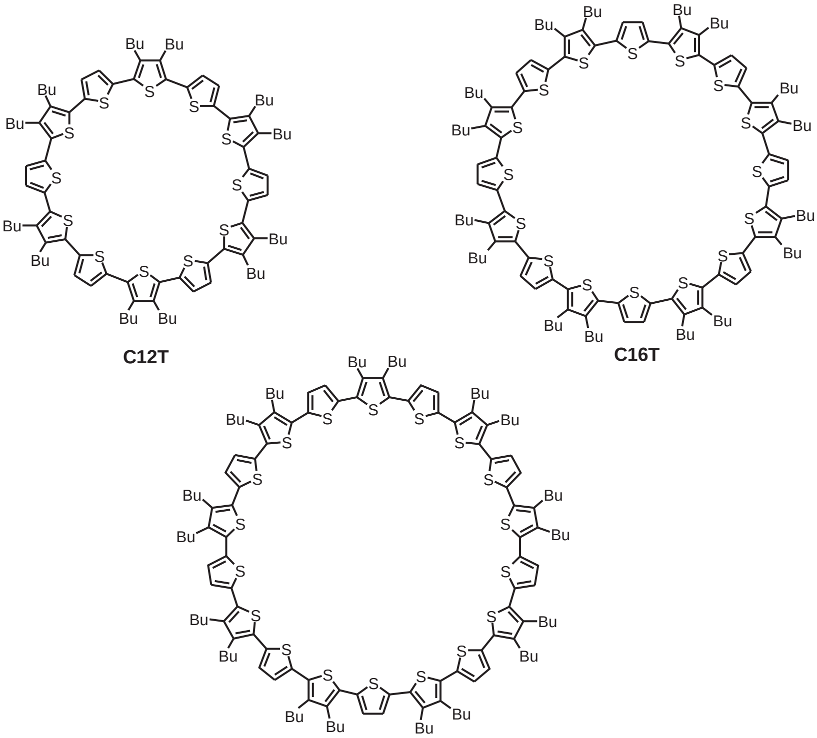
C12T C16T C18T

Fig. 2. Derivatives comprising 12 (C12T), 16 (C16T) and 18 (C18T) thiophene units.

Their optical properties have been investigated in dichloromethane solutions. Absorption spectra of the macrocycles are shown in Fig. 3 and the data summarized in Table 2.

At first glance, we note that in analogy to the linear series the π–π^* transition band is redshifted and becomes progressively more intensive with increasing ring size. However, in comparison to the linear parent compounds with about the same number of thiophene repeating units, these bands exhibit about the same intensity and transition probability, but are strongly shifted to higher energies. E.g., the cyclic 18-mer C18T absorbs at 2.959 eV ($\varepsilon = 123,900$) whereas the linear 19T absorbs at 2.883 eV ($\varepsilon = 125,000$). The transition energy of C18T (2.959 eV) rather corresponds to the one of the linear 9T (2.966 eV, $\varepsilon = 55,700$) but shows about double intensity. The trend that the absorption maxima of the macrocycles are located approximately at energies where linear compounds with half the number of repeating units absorb is also true for the smaller macrocycles and is visualized in Fig. 3 where we implemented the absorption maxima of the linear parent compounds 5T, 7T and 9T as bars. This phenomenon will be explained by our theoretical calculations.

3. Frenkel excitons—influence of geometry

We model the photophysics of oligothiophenes and cyclothiophenes as N optically active two-level monomers forming a simple one-dimensional open linear chain in the case of oligothiophenes and a cyclic chain in the case of cyclothiophenes. It is well known that excitonic methods are appropriate to study conjugated oligomers because it was confirmed that the linear optical absorption is mainly dominated by excitonic states [8]. In our approach we will use as basic excitation a

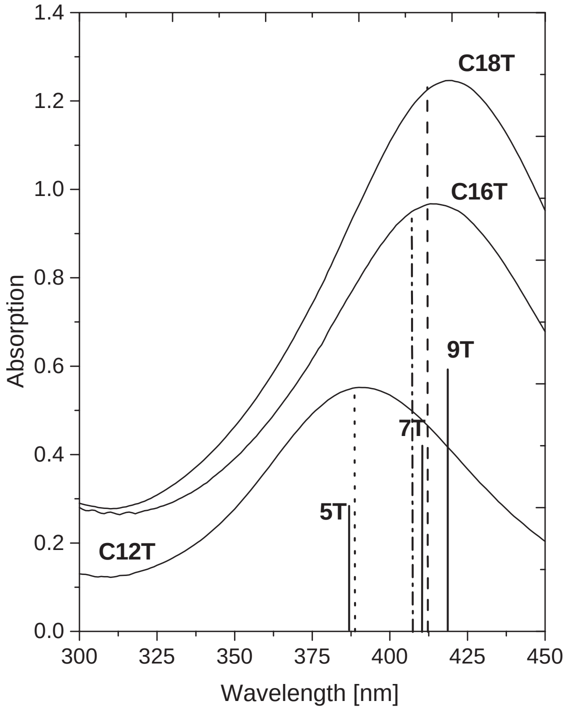

Fig. 3. Solid lines represent experimental absorption spectra for cyclothiophenes, as labeled, with 12, 16, and 18 thiophene units. Solid vertical lines indicate experimental peak position of the corresponding oligothiophenes, labeled by 5T, 7T, and 9T. Dotted, dash-dotted, and dashed vertical lines are the first allowed transition for the cyclothiophenes calculated from Eq. (6) for $N_r = 12, 16,$ and 18 respectively. ($\omega_0 = 5.5\,\text{eV}$, $J = 1.33\,\text{eV}$ as in Fig. 4).

Table 2
Absorption maxima and extinction coefficients of the longest wavelength absorption for cyclo[n]thiophenes CnT

CnT	C12T	C14T	C18T
$\lambda_{\max}^{\pi-\pi^*}$			
(nm)	391	415	419
(eV)	3.171	2.987	2.959
ε(l/mol cm)	55,200	96,700	123,900

collective Frenkel exciton which is the superposition of local excitations of monomers defined by the thiophene units. The corresponding Frenkel exciton Hamiltonian reads

$$H_{\text{ex}} = \omega_0 \sum_{n=1}^{N} |n\rangle\langle n| - J \sum_{n=1}^{N-1} (|n\rangle\langle n+1| + |n+1\rangle\langle n|), \tag{1}$$

where ω_0 is the excitation energy of a monomer and $|n\rangle$ denotes the state vector of the local excitation at the nth monomer. J describes the nearest-neighbor coupling and the sign of J depends on the angle between the transition dipole moment and the chain axis. In our calculations we assume J to be *positive* and thus the states coupling to the light are those at the bottom of the exciton band. For the assumed geometry, Hamiltonian (1) can be diagonalized using the new basis $|k\rangle = \sum_n \phi_k(n)|n\rangle$. The definition of $\phi_k(n)$ for the linear and cyclic geometrics is given by comparison with Eq. (2) and Eq. (5), respectively. For an open chain of length N_l a suitable basis is

$$|k\rangle = \left(\frac{2}{N_l+1}\right)^{1/2} \sum_{n=1}^{N_l} \sin\left(\frac{\pi k n}{N_l+1}\right) |n\rangle, \tag{2}$$

and the corresponding eigenvalues are

$$E_k^l = \omega_0 - 2J\cos\left(\frac{\pi k}{N_l+1}\right). \tag{3}$$

Here, $k = 1, 2, \ldots, N_l$. The state with $k = 1$ lies at the bottom of the exciton band.

In this simple picture, the absorption spectrum shows a series of peaks at the exciton eigenfrequencies, with weights given by the oscillator strengths $O_k = |\sum_n \phi_k(n)\mu_n|^2$ (μ_n is the transition dipole moment at site n, the polarization vector of the light and average over all orientations of the oligomers are omitted). For the open chain, with transition dipole moments at each monomer unit equal in magnitude and parallel to each other, $\mu_n = \mu$, the oscillator strength takes the form

$$O_k^l = \frac{1-(-1)^k}{2} \frac{2\mu^2}{N_l+1} \cot^2 \frac{\pi k}{2(N_l+1)}. \tag{4}$$

In this case only odd states have an oscillator strength, while even ones are dark. Most of the oscillator strength is accumulated in the lower state ($O_{k=1} \approx 0.81(N_l+1)\mu^2$ for a long chain [9]).

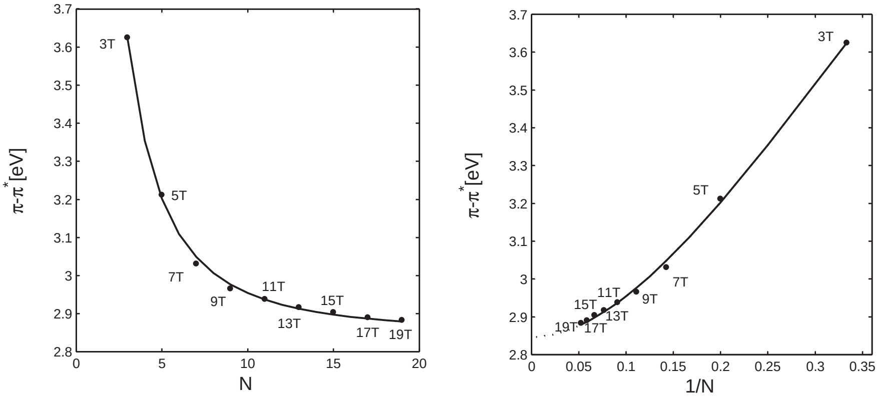

Fig. 4. The left panel shows the dependence of the maximum of the π–π* absorption band on the size of the oligothiophenes (nT, with n the number of thiophene rings). The right panel shows the same photophysical property presented as a function of the inverse chain length $1/N$ to better show saturation for longer chains. The dots mark the experimental data taken from the literature [1–3], while the lines connecting the dots represent theoretical curves based on Eq. (3) ($k = 1$) with fittings parameters $\omega_0 = 5.5\,\text{eV}$ and $J = 1.33\,\text{eV}$. The dotted line on the right panel indicates predictions of the energy gap in the case of an infinite polymer, and the corresponding value is 2.85 eV.

Consequently, we can assume that absorption maxima of oligothiophenes are dominated by the optically allowed transition to the lowest state ($k = 1$) in the excitonic band. Fig. 4, left panel, shows the position of absorption maxima for the series of oligothiophenes in solution at room temperature, experimental points are indicated by large dots labeled nT (n is the number of thiophene units). Using Eq. (3) ($k = 1$), for the parameters $\omega_0 = 5.5\,\text{eV}$ and $J = 1.3\,\text{eV}$ we could easily fit the experimental points. In our model the saturation of the position of the absorption maxima with increasing chain length is explained only by the length-dependent exciton bandwidth, the delocalization length of the excitation is assumed to be identical with the chain length. The right panel shows the same data only presented as a function of the inverse chain length $1/N$ to better demonstrate the saturation for longer chains. The dotted line on the right panel indicates predictions of the energy gap approaching an infinite polymer with the limit 2.85 eV. Eq. (3) describes the band width in the nearest-neighbor coupling approximation. In the case of other substituted oligothiophenes, like described in Ref. [4], it can be too simple and further interactions like long-range dipole–dipole interactions have to be taken into account in order to understand size-dependent effects observed in the absorption spectrum.

For the chain with ring symmetry (linear chain with periodic boundary conditions) and length N_r the one-exciton eigenstates of Hamiltonian (1) are Bloch states along the ring with wave numbers $\frac{2\pi}{N_r}k$, where $k = 0, \pm 1, \pm 2, \ldots, \pm(N_r/2 - 1), N_r/2$ (N_r is taken even).

$$|k\rangle = \frac{1}{\sqrt{N_r}} \sum_{n=1}^{N_r} \exp^{ikn\frac{2\pi}{N_r}} |n\rangle. \tag{5}$$

The eigenstates for the pair $\pm k$ are degenerate, with energy

$$E_k^r = \omega_0 - 2J \cos\left(\frac{2\pi k}{N_r}\right). \tag{6}$$

It is worth to mention at this point that for the open chain the energy of the lowest state is always N_l dependent, but as it is easily seen from the last equation this is not the case for the ring shaped oligothiophenes the lowest state has always energy $E_{k=0}^r = \omega_0 - 2J$.

In order to fix the molecular transition dipoles, we assume that all monomers have dipoles that are equal in magnitude (μ) like in the case of an open chain. The components of the dipoles in the direction parallel to z-axis (perpendicular to the molecular ring plane) are equal for all monomers: $\mu_\parallel = \mu \sin\beta$, with β the angle between the monomer dipoles and the plane of the ring; the components in the plain have magnitude $\mu_\perp = \mu \cos\beta$ and rotate around the ring (perpendicular to the z-axis). Arbitrarily assigning the x direction to the in-plane dipole component of the monomer $n = 1$ and the y direction perpendicular to it in the ring plane, the transition dipole of the molecule on site n reads

$$\vec{\mu}_n = \left(\mu_\perp \cos\left(\frac{2\pi n}{N_\mathrm{r}}\right), \mu_\perp \sin\left(\frac{2\pi n}{N_\mathrm{r}}\right), \mu_\parallel\right), \tag{7}$$

(arrows over symbols denote three-dimensional vectors in real space). The oscillator strength thus takes the form

$$O_k^\mathrm{r} = N_\mathrm{r}\mu^2(\sin^2\beta\delta_{k,0} + \tfrac{1}{2}\cos^2\beta(\delta_{k,-1} + \delta_{k,1})). \tag{8}$$

Thus, only three states are dipole allowed, namely $|k\rangle = |0\rangle, |\pm 1\rangle$ (for details see Ref. [10]). We note that the latter two states are degenerate (cf. Eq. (6)) with the polarization of the transition perpendicular to the ring z-axis, while the first transition is polarized parallel to it. In both cases, i.e. for open and closed chains, the oscillator strengths scale with chain length.

Finally, we can show basic consequences of our comparative study. Assuming that for the thiophenes with the circular geometry all transition dipoles are in the ring plane (no z component), the oscillator strength is located only in the first excited state. From Eq. (6) one can immediately predict the position of the absorption maxima of the cyclothiophenes. According to this expression, for $k = \pm 1$ and with ω_0 and J values as found for the thiophene oligomers we can get the predicted peak positions for the synthesised cyclothiophenes with $N_\mathrm{r} = 12$, 16, and 18 repeat units at 3.196, 3.043, and 3.004 eV, respectively. The experimental peak maxima are at 3.171, 2.987, and 2.959 eV, respectively (see Table 2 and Fig. 3). Our estimation and experimental points agree very well. We also can establish an analytical relation between the position of the transitions of the states which mainly contribute to the absorption spectrum in the case of linear and cyclic thiophenes. By equating Eqs. (3) with $k = 1$ and (6) with $k = \pm 1$ we immediately have the relation

$$N_\mathrm{r} = 2(N_\mathrm{l} + 1). \tag{9}$$

The last relation and the previous observations that the oscillator strength scales linearly with N (see Eqs. (4) and (8)) leads to the fact that also in the absorption intensity ratio between linear and cyclic thiophene chains a numerical factor 2 should be seen. Indeed extinctions coefficients confirm this prediction very well (see Table 1 and 2). Instead put "First, as predicted according to the oscillator strength scaling, this coefficients scales linearly with number of repeat units. For example, the oligothiophenes chain with 15 repeat unit shows five (three) times larger extinction coefficient than 3T (5T) (see Tabel 1).". Second, the factor two is very clearly seen in a study comparing the absorption spectra, for example, oligothophenes with 5 repeat units (extinction = 26,600[1/mol cm]) very nicely overlap with cyclothiophenes with 12 repeat units and two times larger.

4. Summary and concluding remarks

Conjugated oligothiophenes and corresponding cyclothiophenes both show size-dependent effects in the photophysical properties which can be described and understood in the frame of a Frenkel exciton theory. The main differences in the observed absorption spectra of oligomers and cyclothiophenes are caused by different selection rules as a consequence of their different geometry. The saturation of the absorption maxima for the longer chains is explained by the size-dependent exciton band width. The conjugation length is assumed to be equal to the chain length. Cyclic thiophenes have the transition dipoles mainly arranged in the ring plane with a very small perpendicular component.

Acknowledgements

The authors gratefully acknowledge the Deutsche Forschungsgemeinschaft for financial support within a SFB 569.

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